# Lecture 13

## Hedonics and Real Estate Markets

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ECON 467

# Roadmap

- What can we use to infer the demand for environmental goods?
- What do housing prices tell us?
- What is the demand for hazardous waste? (Greenstone and Gallagher, 2008)
- What is the demand for sea level rise? (Bernstein, et al. 2019)

# Hedonic valuation

# Revealed preference approaches

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Is there a way we can reveal the value of these goods?

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This change in price can tell us something about how people value the change in the environmental good

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What does this price change mean?

# Hedonics: Property value models

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- Rooms
- Bathrooms
- School quality
- **Environmental quality**

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Homes located in pristine areas are likely to be more valuable than identical homes located near toxic facilities

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Property purchases are large and consequential: buyers and sellers are likely to be well-informed

It is uncontroversial that property values should reflect local attributes

e.g. homes in better school districts are typically more expensive

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By 2005: $35 billion in federal funding has been spent at roughly 800 sites

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How do we do it?

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Main idea: Take two otherwise very similar houses: one in a neighborhood surrounding a site that has been cleaned up and one in a neighborhood surrounding a site that has not

How do their prices differ?

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Ridker and Henning (1967) first applied method to environmental valuation in a study of the effect of air pollution on property values in St. Louis

It motivated the conceptual model of Rosen (1974) of how we might use hedonic prices to estimate peoples' values for site-specific amenities

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Lets get some intuition for how housing markets reveal the value of environmental goods

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At the current equilibrium price of $200,000 per house, all 200 hundred homes on either lake are equally preferred

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Lake A prices **increase** to bring the market back into equilibrium

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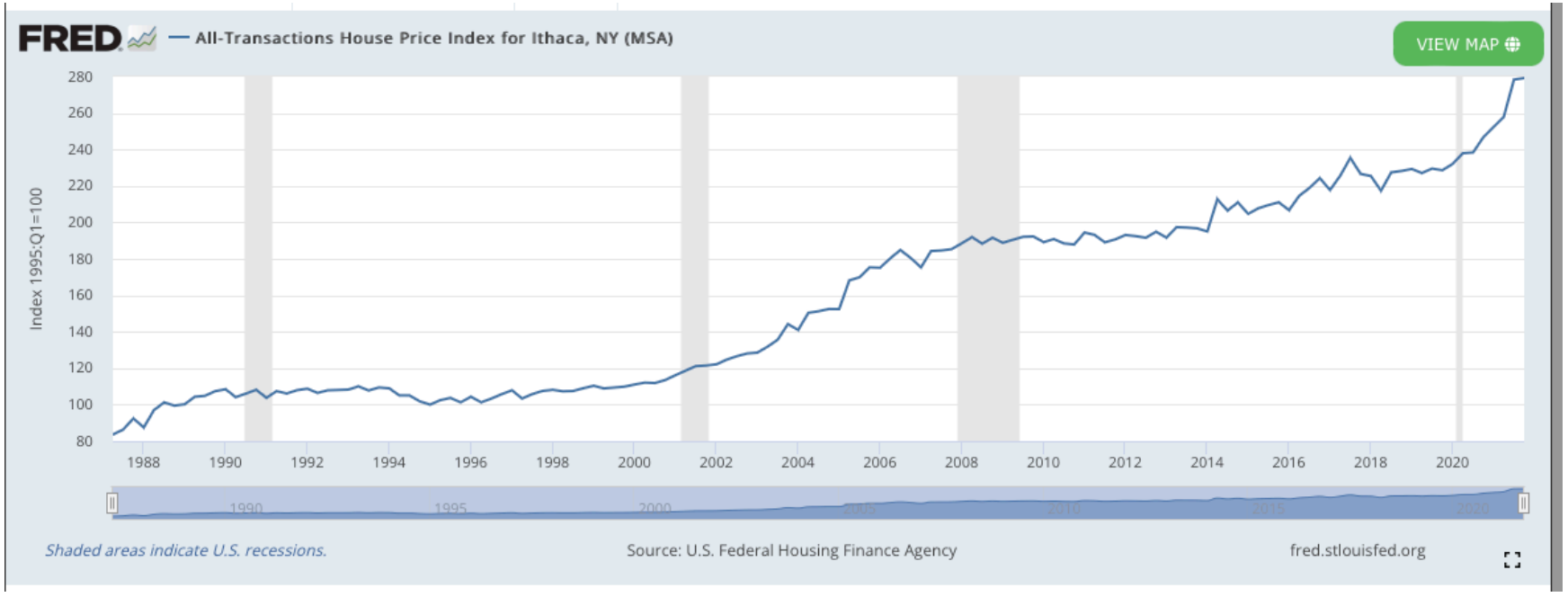
**Sidebar:** think about US cities in the last 20 years and urban residential

# Housing prices in Ithaca are increasing **fast**, why?

## Study shows Ithaca home prices rising far faster than nation's



# Another Dyson professor's house

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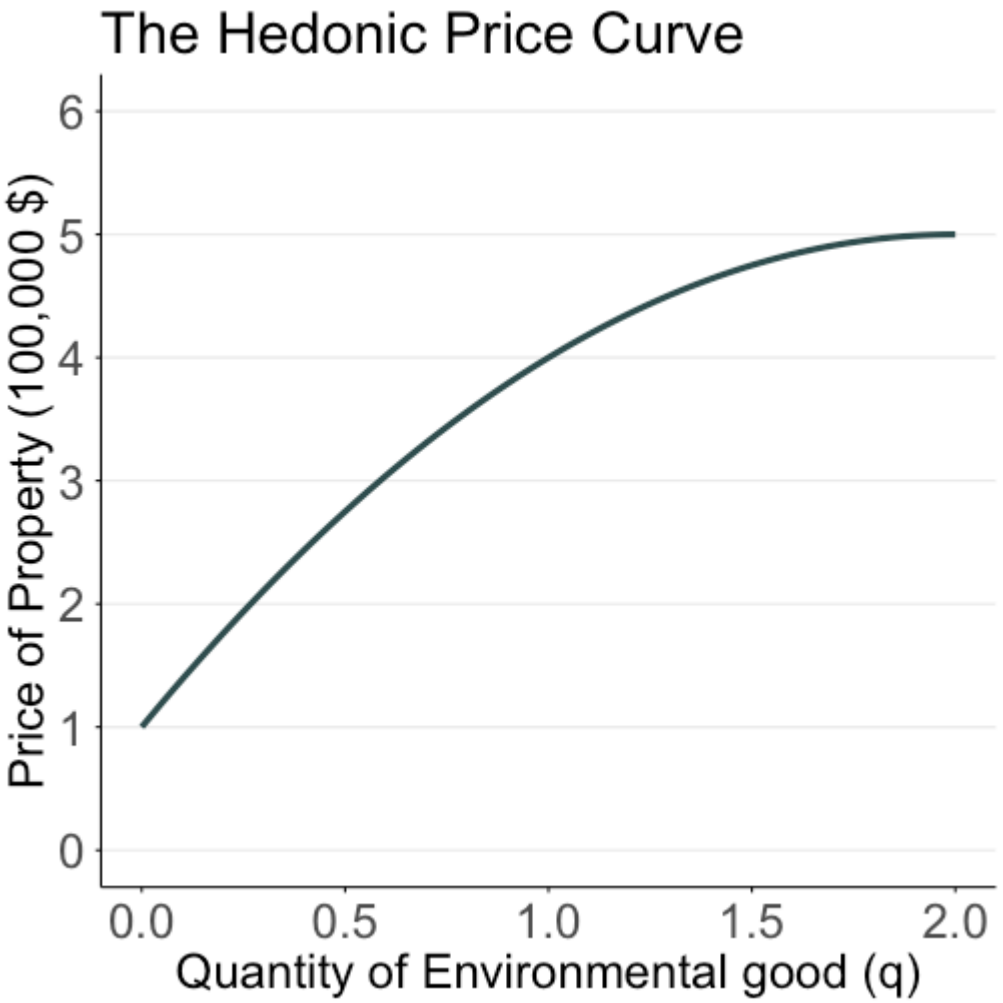
Here we will assume the supply of houses is fixed in the short run so the price curve arises solely from buyer behavior

# The hedonic model: the price curve



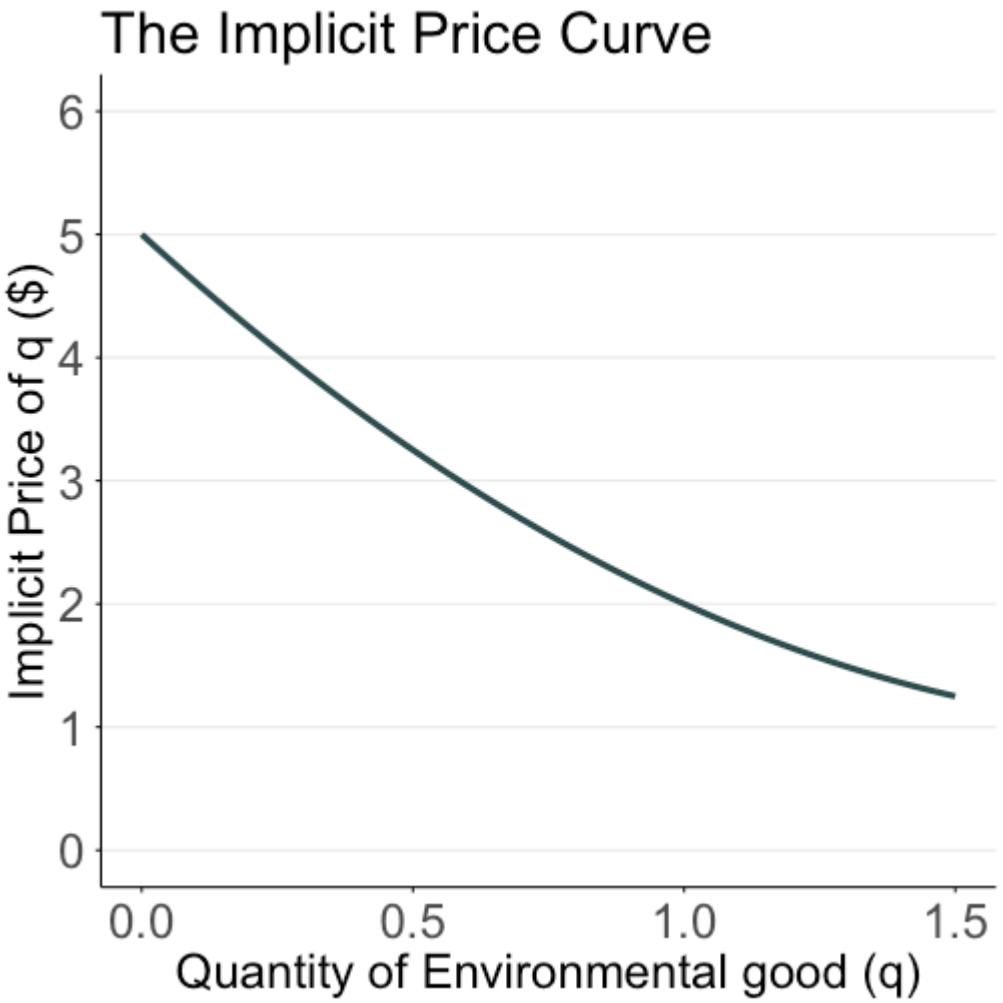

The hedonic price curve is $P(x, q)$

It's increasing in q (q is good) but at a decreasing rate (decreasing marginal utility)

This is holding $x$ fixed

Analogous to regular demand curves holding income fixed

# The hedonic model: the price curve




The implicit price curve for q is

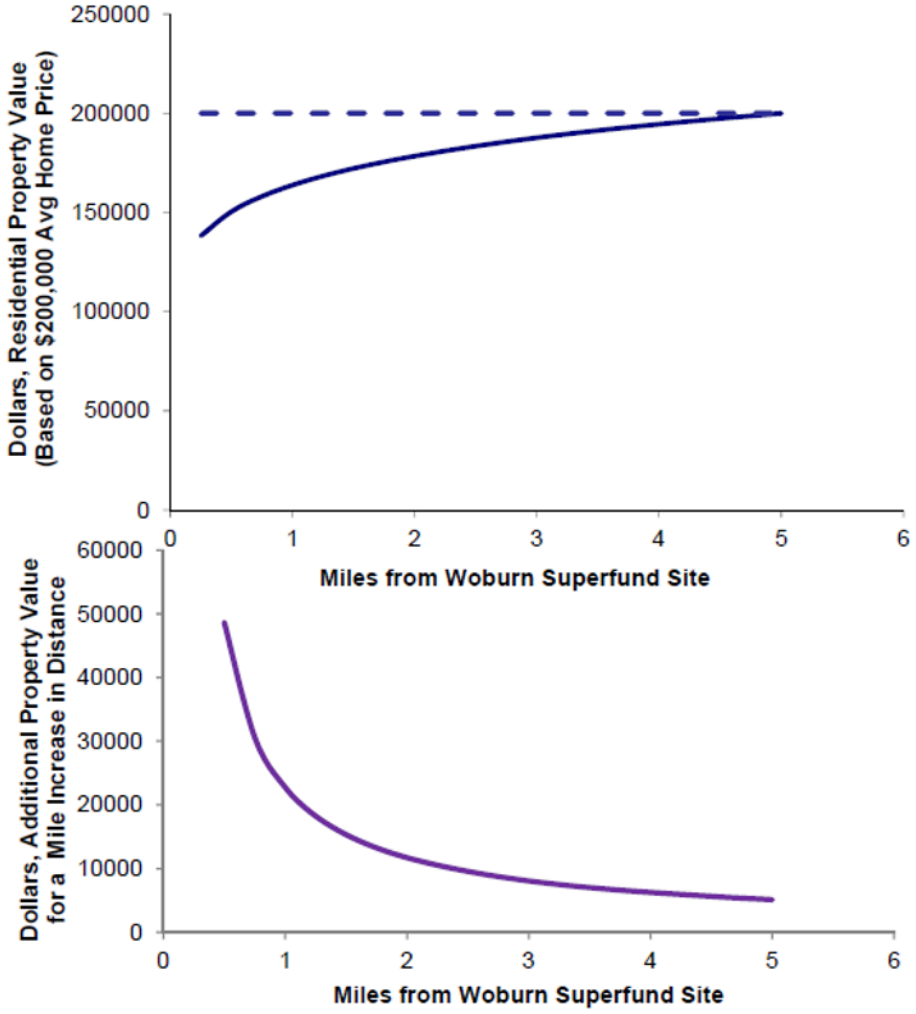
$$\frac{\partial P(x,q)}{\partial q}$$

It tells us how the price changes in q

It's positive, but downward sloping

This is effectively the environmental good demand curve

# Price curve example




Total Value

"Marginal" Value
(one mile increment)

Source: Messer et al. *Env. and Res. Econ.* 2006

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- $z$ is the numeraire good (spending on other private goods)
- $y$ is income
- $s$ is the set of the household's characteristics like family size, ages, etc

# Unrealistic pieces of the model

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Another is that you just can't purchase some sets of $x$ (i.e. a huge lot in downtown Manhattan with a farm)

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Many housing characteristics are discrete (bedrooms, bathrooms, etc)

Another is that you just can't purchase some sets of $x$ (i.e. a huge lot in downtown Manhattan with a farm)

We won't touch on this in class because it's a lot more complicated, but economists know how to deal with these problems

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We are thus also implicitly assuming $q$ varies across space so that households can sort into areas they prefer

- q is really picking up **local** environmental goods

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For homeowners we are basically assuming they rent from themselves every year

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$$\max_{x,q,z} U(x, q, z; s) \quad \text{subject to:} \quad y = z + P(x, q)$$

Plug in the constraint for $z$ to get:

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The FOCs for this problem are:

$$\frac{\partial U}{\partial x_j} = \frac{\partial U}{\partial z}\frac{\partial P}{\partial x_j} \quad j = 1, \ldots, J \quad \text{(house characteristics)}$$

$$\frac{\partial U}{\partial q} = \frac{\partial U}{\partial z}\frac{\partial P}{\partial q} \quad \text{(environmental good)}$$

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What does this mean?

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**Recall from intro/intermediate micro:** the MRS tells us how the household trades off $q$ and $z$ while keeping utility constant

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How prices change in the environmental good, *holding all else constant*, tells us about WTP

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e.g. what if air quality improved in Syracuse because we are in a recession?

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- But recessions also decrease demand for houses and make prices lower (people are unemployed)

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The housing prices went up **despite** the recession!

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The change in prices of Ithaca houses tells us the impact of the recession, if we subtract it from the change in prices of Syracuse homes we get the effect of air quality alone!

- Syracuse price change - ithaca price change = air quality effect
- (air quality effect + recession effect) - (recession effect) = air quality effect

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# Housing prices and superfund clean up

Greenstone, Michael, and Justin Gallagher (2008). “Does Hazardous Waste Matter? Evidence from the Housing Market and the Superfund Program.” Quarterly Journal of Economics

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**Main question:** How does superfund site clean up affects the housing price in the adjacent areas?

# Housing prices and superfund clean up

Greenstone, Michael, and Justin Gallagher (2008). “Does Hazardous Waste Matter? Evidence from the Housing Market and the Superfund Program.” Quarterly Journal of Economics

**Main question:** How does superfund site clean up affects the housing price in the adjacent areas?

**How they do it:** Compare housing market outcomes in the areas surrounding the first 400 hazardous sites chosen for Superfund clean-ups to the areas surrounding the 290 sites that narrowly missed qualifying for these clean-ups

# Housing prices and superfund clean up

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**Key idea:** Any differences between housing values in these locations is most likely due to Superfund clean up, not other factors

# Superfund location

Figure IIa
GEOGRAPHIC DISTRIBUTION OF HAZARDOUS WASTE SITES IN THE 1982 HRS SAMPLE SITES WITH 1982 HRS SCORES EXCEEDING 28.5

Figure IIb
GEOGRAPHIC DISTRIBUTION OF HAZARDOUS WASTE SITES IN THE 1982 HRS SAMPLE SITES WITH 1982 HRS SCORES BELOW 28.5

# Regression

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$$\log(2000 \text{ median home price})_c = \theta \underbrace{1(\text{cleaned up in } 2000)_c}_{=1 \text{ if true}, =0 \text{ otherwise}} + \beta \underbrace{X_c}_{\text{controls}} + \varepsilon_c$$

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They are interested in $\theta$ which tells us the percent change in a census tract median home price if it was cleaned up

$-\theta$ is telling us the **cost** of a superfund site to households

# Superfund results: "quasi-experimental"

QUASI-EXPERIMENTAL ESTIMATES OF THE EFFECT OF NPL STATUS ON HOUSE PRICES, SAMPLES BASED ON THE 1982 HRS SAMPLE SITES

| | | | | | RD-Style Estimators | | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| | A. Own Census Tract | | | | | | |
| 1(NPL Status by 2000) | 0.035 | 0.037 | 0.043 | 0.047 | 0.007 | 0.022 | 0.027 |
| | (0.031) | (0.035) | (0.031) | (0.027) | (0.063) | (0.042) | (0.038) |
| | B. Adjacent Census Tracts | | | | | | |
| 1(NPL Status by 2000) | 0.071 | 0.066 | 0.012 | 0.015 | -0.006 | -0.002 | 0.001 |
| | (0.031) | (0.035) | (0.029) | (0.022) | (0.056) | (0.035) | (0.035) |
| | C. 2-Mile Radius from Hazardous Waste Sites | | | | | | |
| 1(NPL Status by 2000) | 0.021 | 0.019 | 0.011 | 0.001 | 0.023 | -0.018 | -0.007 |
| | (0.028) | (0.032) | (0.029) | (0.023) | (0.054) | (0.035) | (0.034) |
| Ho: > 0.138, P-Value | 0.000 | 0.000 | 0.000 | 0.000 | 0.018 | 0.000 | 0.000 |

Top row of the last three columns are the important ones

# Superfund results

Superfund cleanups had **economically and statistically insignificant effects** on property values, rental rates, housing supply, population, who lives near the site: 0.7-2.7% depending on the model

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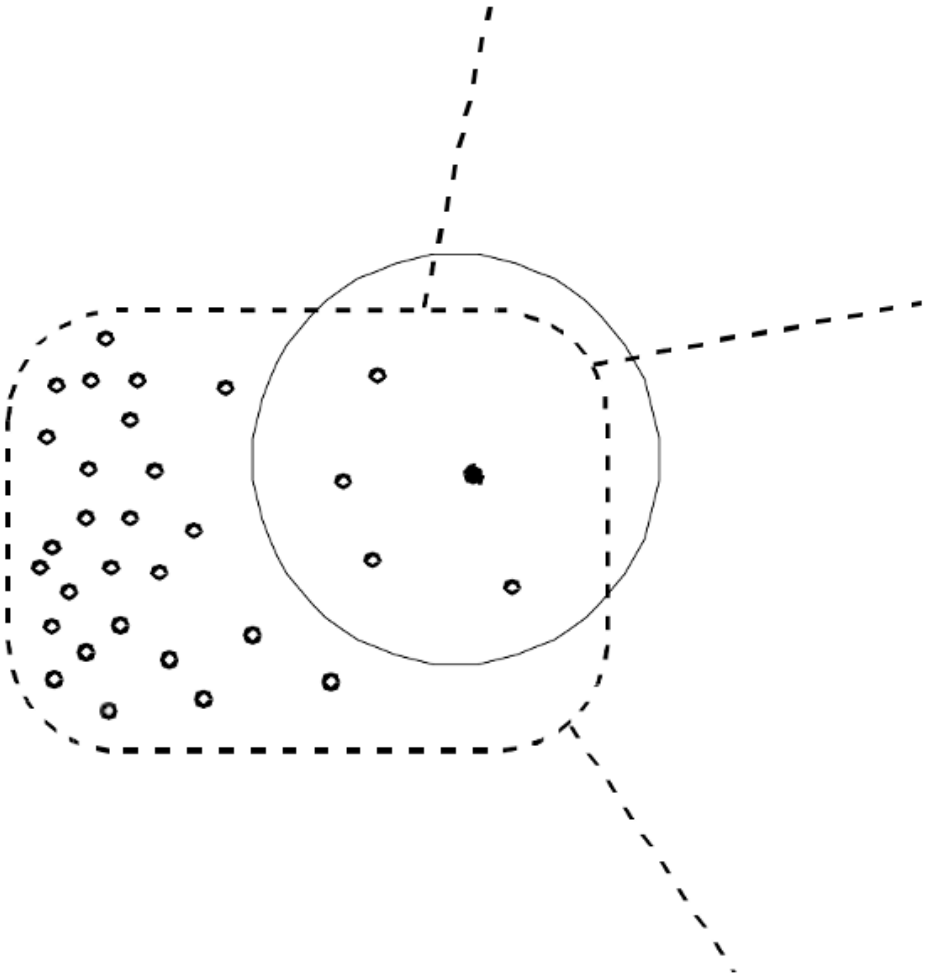
Why does granularity matter?

# Superfund: zoom in

Superfund sites are a localized disamenity

Previous attempts to value cleanup looked at changes in census tract median housing values and found no impacts

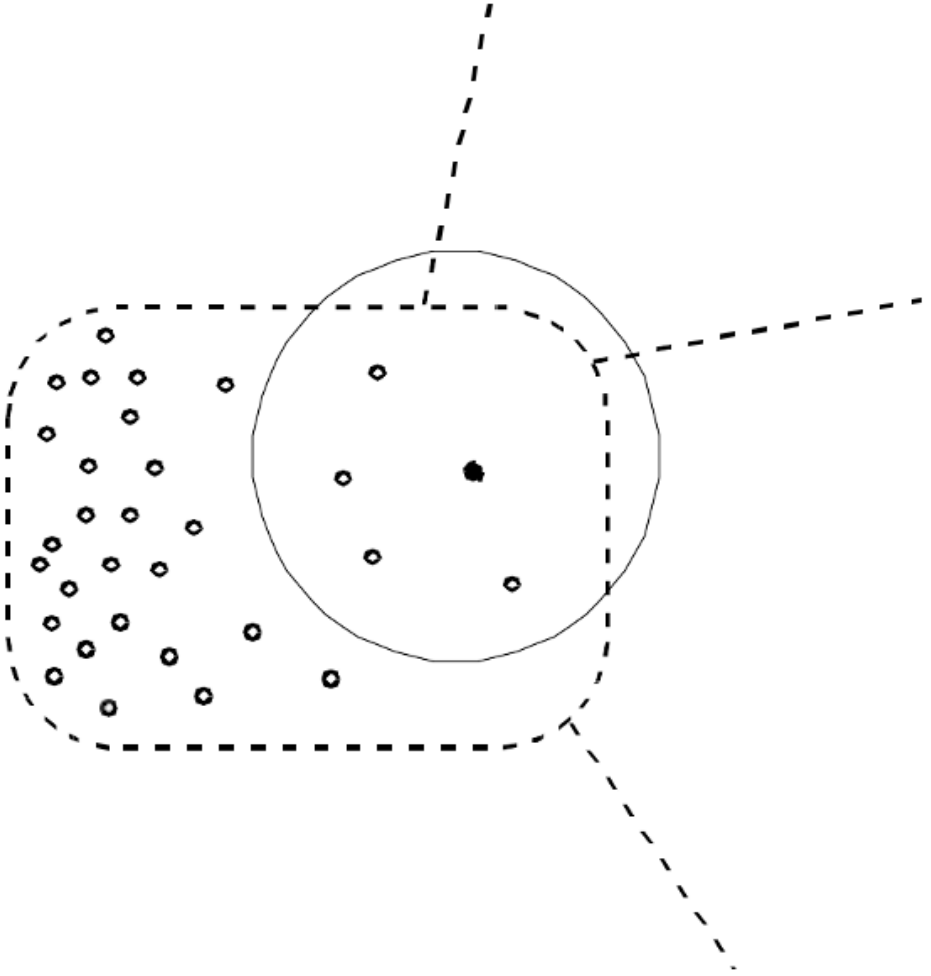
Need to look **within** census tracts



# Superfund: zoom in

Consider changes in other percentiles of within-tract house value distribution:

*deletion of a site raises tract-level housing values by 18.2% at the 10th percentile, 15.4% at the median, and 11.4% at the 60th percentile*



# Sea level rise

Sea level rise (SLR) is a long run phenomenon

Not **a lot** of flooding now, but by 2050 sizable portions of NYC will be flooded

By 2100 average SLR will be over 2 feet

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**Yes,** why?

Let's work through the logical steps

# Sea level rise

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Effects decades in the future can affect current prices

# Sea level rise

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Houses are kind of like annuities:

- Pay an upfront cost (mortgage)
- Get a future stream of revenues (rental payments from renters)

The price of an annuity should be equal to the present value of the stream of payments (minus upkeep costs)

- Think about why this must be true

# Sea level rise

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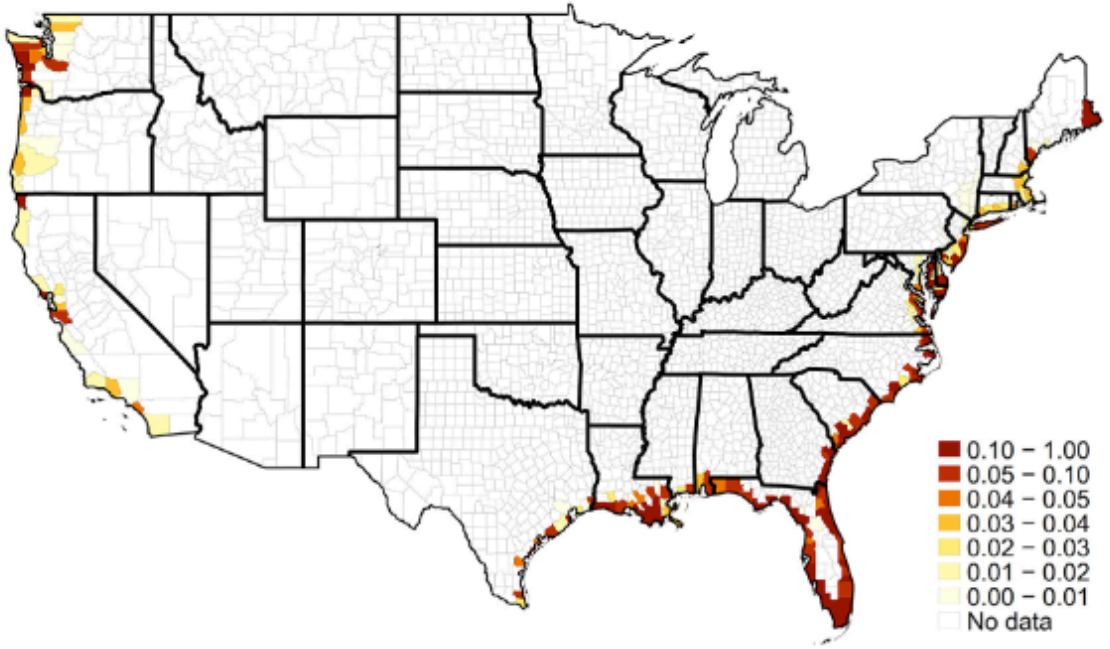
If SLR reduces future demand for rentals (decreases rental payments) or increases upkeep costs (e.g. more maintenance of the house), future rental profit goes down

Similar to annuities, this should decrease the price of the house

# Sea level rise: where is it happening?

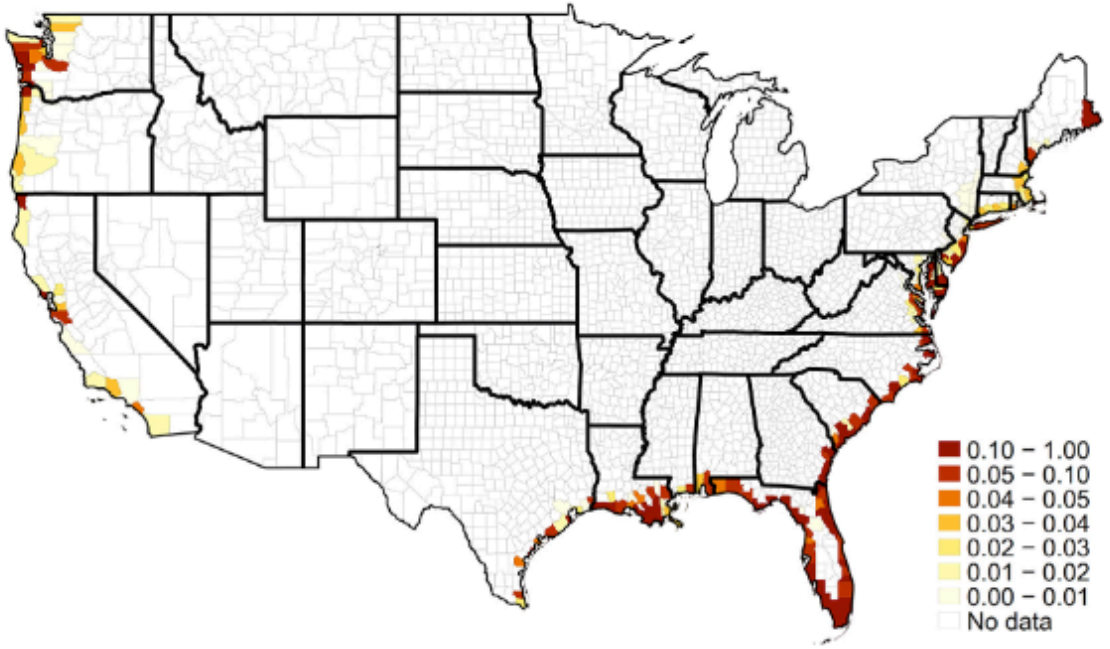
The map shows the share of houses sold between 2007-2017 that would be flooded with 6 feet of SLR



# Sea level rise: where is it happening?

The map shows the share of houses sold between 2007-2017 that would be flooded with 6 feet of SLR

Lots of houses in the Southeast are exposed!



# Sea level rise and housing prices

Bernstein, Gustafson, and Lewis (BGL) (2019) estimate how expected SLR affects current housing prices

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How do they do it?

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How do they do it?

Use a regression model to compare houses exposed to different amounts of SLR, but controlling for (i.e. have the exact same):

- Distance to the coast
- Zipcode
- Property characteristics (bedrooms, bathrooms, square footage, etc)
- Month of sale

# Sea level rise: where is it happening?

BGL are *basically* computing the difference in house prices between two houses that are identical, in the same place, but one happened to be at higher elevation

This zipcode is only 92 square miles, and between 3 and 20 feet of elevation

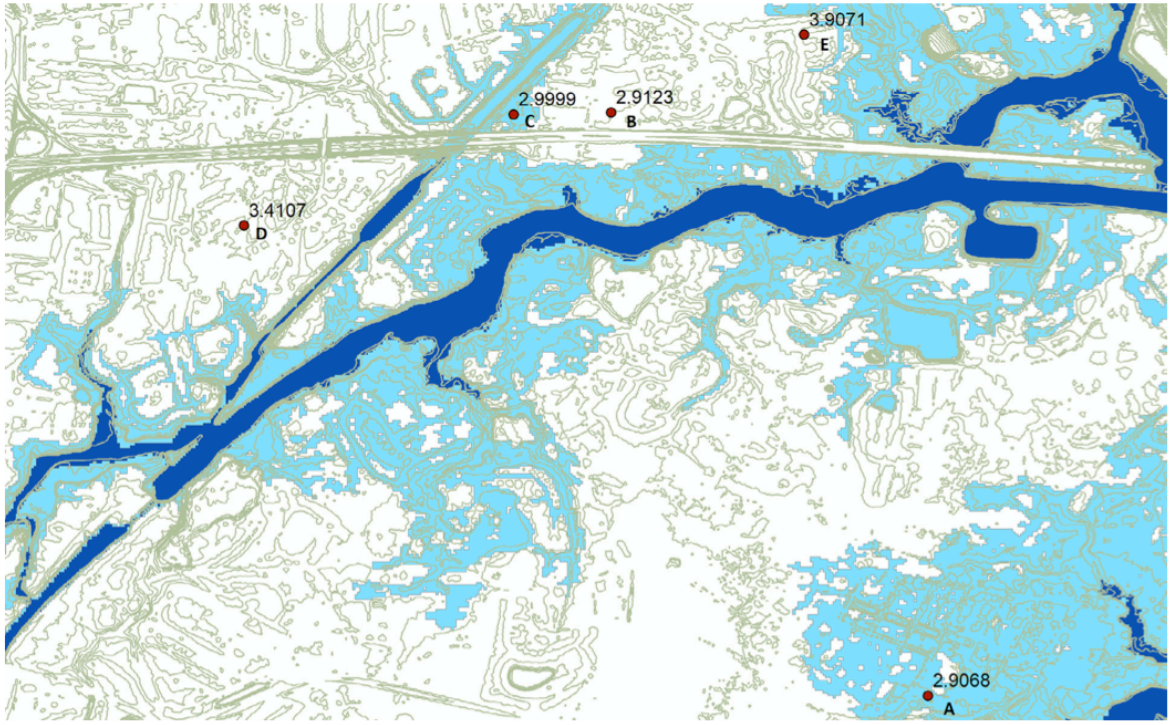
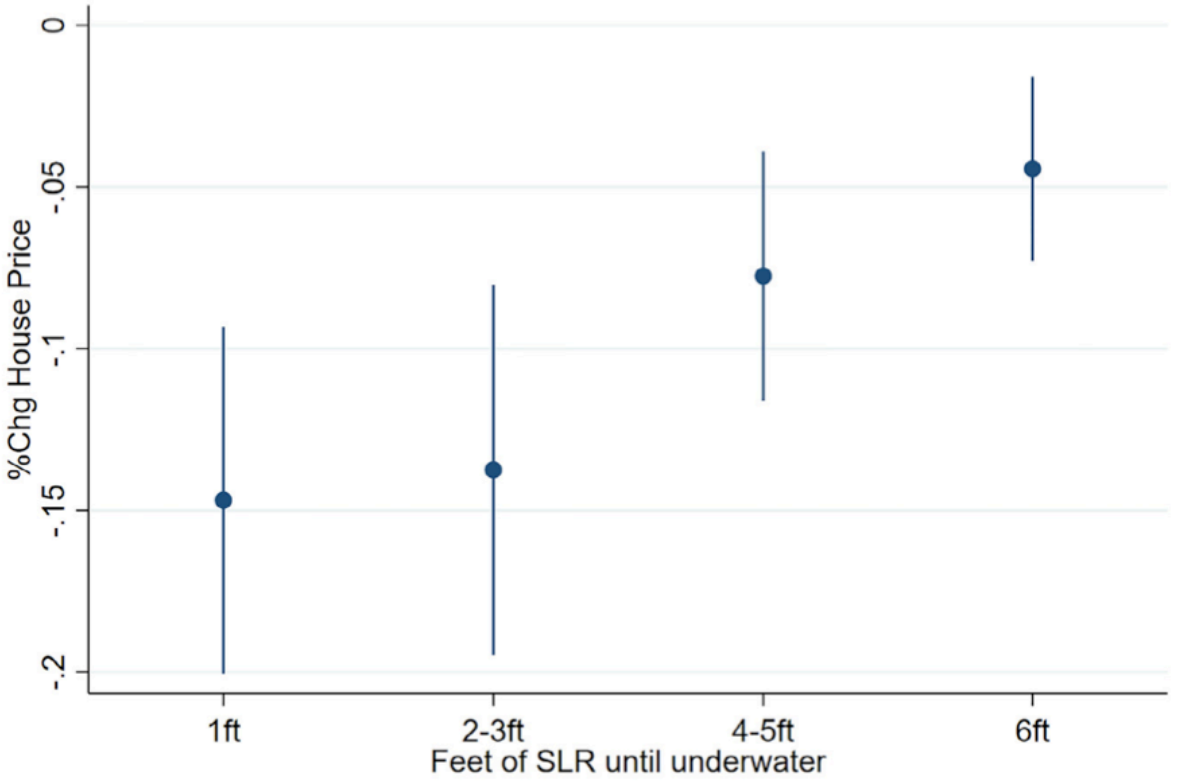



**Fig. 2.** Example of within-bin variation in SLR exposure. Fig. 2 displays five transactions in zip code 23323 (in Chesapeake, VA) during July of 2014, each of which involves a property that is (1) between 0.16 and 0.25 miles from the coast, (2) elevated between two and four meters above sea level, (3) four bedrooms, (4) a non-condominium, (5) owner occupied, (6) bought by a non-local buyer. Properties are labeled A–E, with elevation in meters above the property label. The olive contour lines represent 2-foot elevation contours. The dark blue area is the NOAA zero-foot SLR layer indicating the point of the highest high tide today while the light blue is the 6-foot layer indicating the highest high tide after six feet of global average sea level rise.

# Sea level rise: what is the effect?

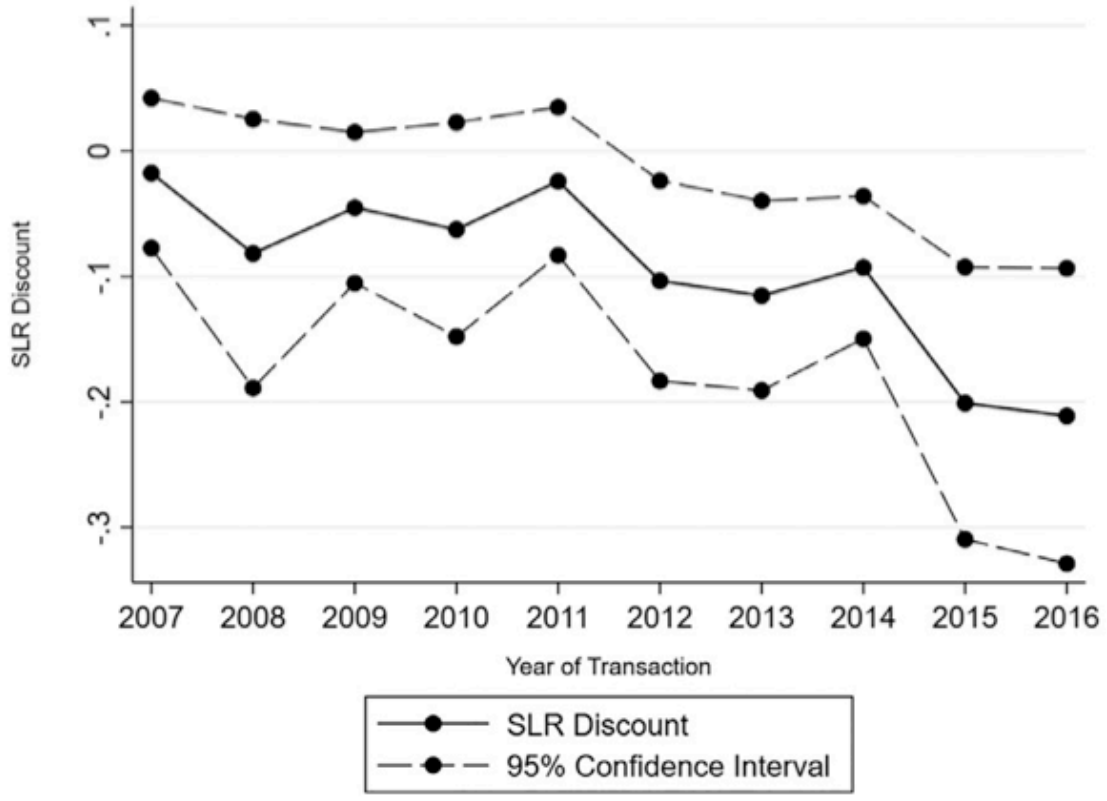
Houses that would be under water with 1 foot of SLR sell **15 percent** cheaper than the exact same house that is not SLR-exposed

The discount for houses exposed to 6 feet of SLR is only 5%



# Sea level rise: what is the effect?

The discount from SLR (>6 feet) is getting **bigger** over time



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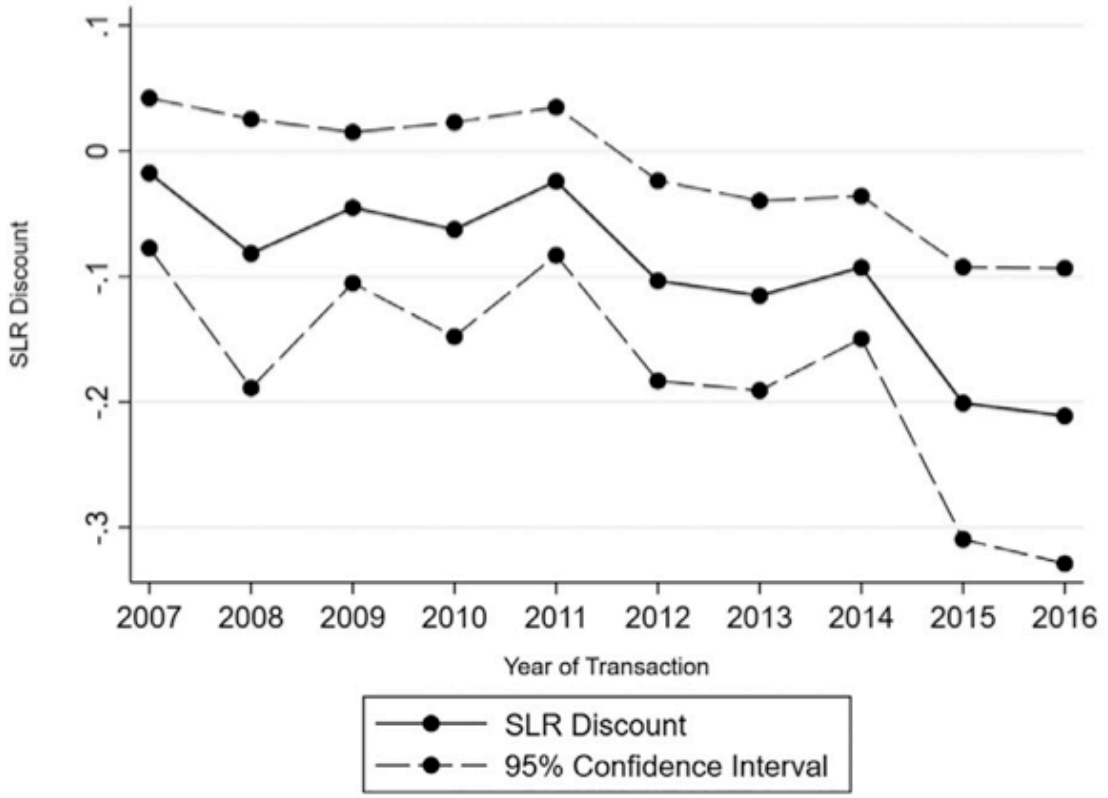

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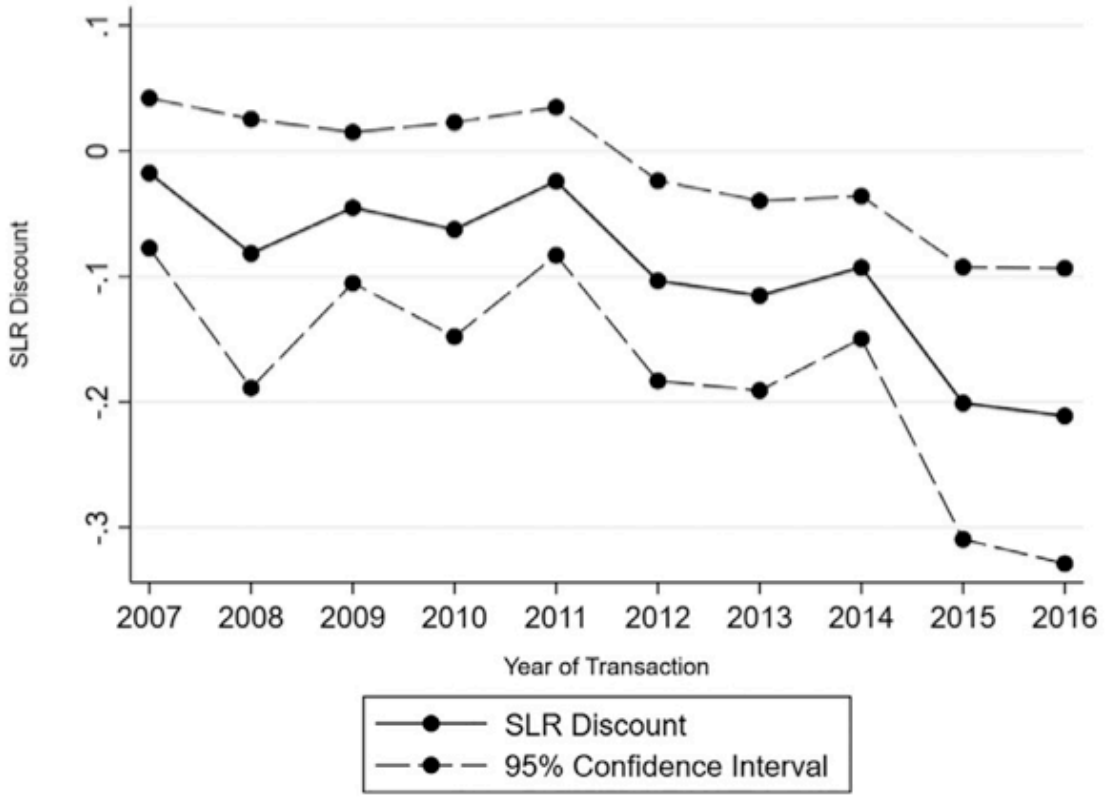
SLR projections may be updated over time and more dire



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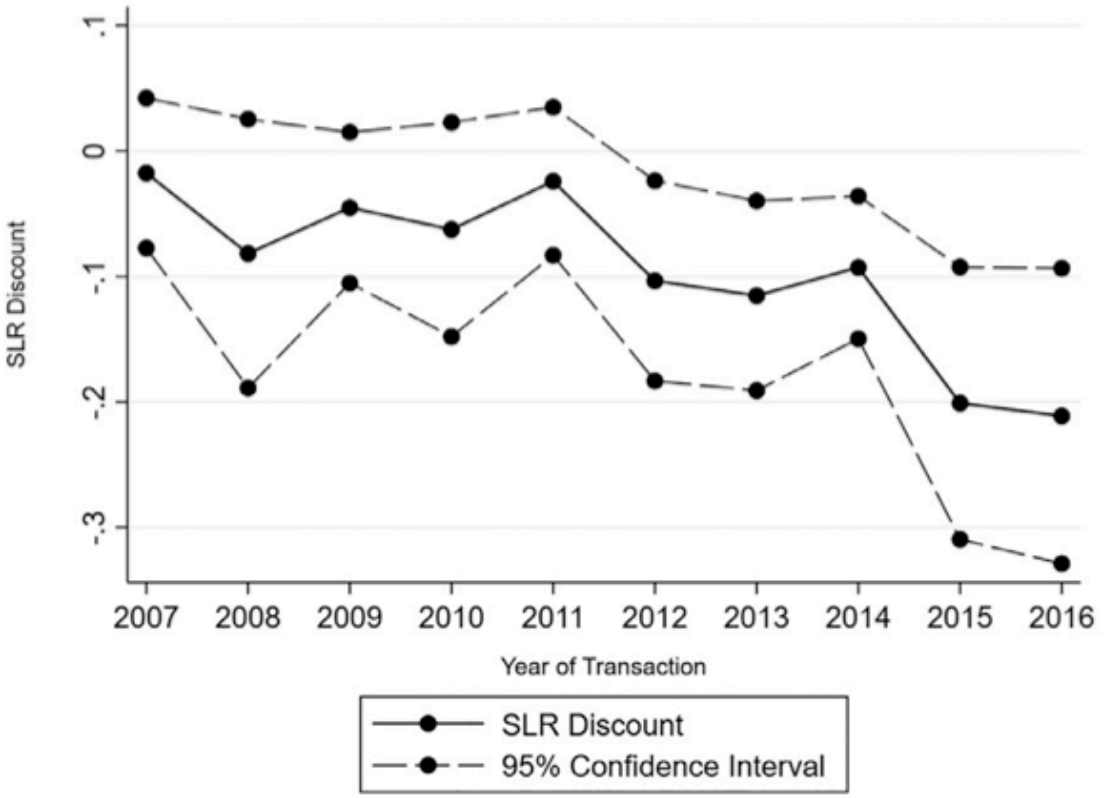
Buyers may be becoming more informed about SLR



# Sea level rise: what about rents?

SLR isn't happening until far into the future so it shouldn't affect rents **today**

| ln(price/sqft) | ln(price) |
|---|---|
| (3) | (4) |
| −0.003 | −0.014 |

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BGL estimate how future SLR affects current rents and finds very small effects like we'd expect: discounts of 1.4% or smaller

# Value of a statistical life (VSL)

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A statistical life is a probabilistic concept

VSL reflects willingness to pay for a reduction in the risk of death

VSL is more appropriately called the **value of mortality risk**

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How?

See tradeoffs people make between cost and safety

# Value of a statistical life (VSL)

Some examples:

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Driving speed

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Wage-risk relationship

There's lots of studies, and lots of different answers

# VSL

> EPA recommends that the central estimate of $7.4 million ($2006), updated to the year of the analysis, be used in all benefits analyses that seek to quantify mortality risk reduction benefits regardless of the age, income, or other population characteristics of the affected population until revised guidance becomes available

# VSL thought experiment

Suppose that individuals are willing to adopt a safety procedure, for which they have to give up 25 cents per hour, to reduce risk of on-the-job fatality by 1 in 10,000 (annual risk)

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VSL = $500*10,000 = 5 million dollars

# Estimating a hedonic wage function

We can estimate a **hedonic wage function:**

$$w_i = \alpha + \beta_1 H_i + \beta_2 X_i + \gamma_1 p_i + \gamma_2 q_i + \gamma_3 WC_i + \varepsilon_i$$

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$w$: wage

$H$: worker personal characteristics

$X$: job characteristics

$p$: **risk of death at the job**

$q$: non-fatal risk at the job

$WC$: workers' compensation benefits for injury

$\frac{\partial w}{\partial p}$ is the wage-risk trade off for marginal changes in risk

# VSL from the hedonic wage function

Suppose:

- Wages were in thousands of dollars
- Risk is deaths per 10,000 people
- Coefficient on mortality risk $p$ is $\gamma_1 = 0.4$

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# VSL from the hedonic wage function

Suppose:

- Wages were in thousands of dollars
- Risk is deaths per 10,000 people
- Coefficient on mortality risk $p$ is $\gamma_1 = 0.4$

This implies an average WTP (reduced wage) of 400 dollars to reduce risk by 1 in 10,000

# VSL from the hedonic wage function

WTP (reduced wage) of 400 dollars to reduce risk by 1/10,000

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Suppose a policy reduces mortality risk by 1/10,000 for 60,000 people (saves 6 lives on average)

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Suppose a policy reduces mortality risk by 1/10,000 for 60,000 people (saves 6 lives on average)

This policy has a value of:

$$400 * 60,000 = 24 \text{ million dollars}$$

# VSL estimates

**Exhibit 7-3** Value of Statistcal Life Estimates (mean values in 1997 dollars)

| Study | Method | Value of Statistical Life |
|---|---|---|
| Kneisner and Leeth (1991 - U.S.) | Labor Market | $0.7 million |
| Smith and Gilbert (1984) | Labor Market | $0.8 million |
| Dillingham (1985) | Labor Market | $1.1 million |
| Butler (1983) | Labor Market | $1.3 million |
| Miller and Guria (1991) | Contingent Valuation | $1.5 million |
| Moore and Viscusi (1988) | Labor Market | $3.0 million |
| Viscusi, Magat and Huber (1991) | Contingent Valuation | $3.3 million |
| Marin and Psacharopoulos (1982) | Labor Market | $3.4 million |
| Gegax et al. (1985) | Contingent Valuation | $4.0 million |
| Kneisner and Leeth (1991 - Australia) | Labor Market | $4.0 million |
| Gerking, de Haan and Schulze (1988) | Contingent Valuation | $4.1 million |
| Cousineau, Lecroix and Girard (1988) | Labor Market | $4.4 million |
| Jones-Lee (1989) | Contingent Valuation | $4.6 million |
| Dillingham (1985) | Labor Market | $4.7 million |
| Viscusi (1978, 1979) | Labor Market | $5.0 million |
| R.S. Smith (1976) | Labor Market | $5.6 million |
| V.K. Smith (1976) | Labor Market | $5.7 million |
| Olson (1981) | Labor Market | $6.3 million |
| Viscusi (1981) | Labor Market | $7.9 million |
| R.S. Smith (1974) | Labor Market | $8.7 million |
| Moore and Viscusi (1988) | Labor Market | $8.8 million |
| Kneisner and Leeth (1991 - Japan) | Labor Market | $9.2 million |
| Herzog and Schlottman (1987) | Labor Market | $11.0 million |
| Leigh and Folsom (1984) | Labor Market | $11.7 million |
| Leigh (1987) | Labor Market | $12.6 million |
| Garen (1988) | Labor Market | $16.3 million |
| Derived from EPA (1997) and Viscusi (1992). | | |